

Musical Instrument E-Commerce Management System

CHAPTER – 2

Introduction

The Amazon Musical Instrument E-Commerce Management System is designed to manage the online sale of musical instruments such as guitars, keyboards, pianos, drums, violins, microphones, amplifiers, and other related accessories. A well-structured database is essential for organizing customer information, seller details,

In this phase of the database design, the main entities, their attributes, primary keys, foreign keys, and relationships are identified. Defining these components helps create an efficient Entity-Relationship (ER) model, reduces data redundancy, maintains data integrity, and ensures smooth business operations. This database design provides a strong foundation for implementing a reliable and scalable Amazon Musical Instrument E-Commerce platform.

2.1 Customer

Attribute	Key	Description
Customer_ID	Primary Key (PK)	Unique ID of the customer
First_Name	-	Customer's first name
Last_Name	-	Customer's last name
Email	-	Customer's email address
Phone	-	Customer's contact number
Address	-	Customer's delivery address
Password	-	Customer's account password

2.2 Seller

Attribute	Key	Description
Seller_ID	Primary Key (PK)	Unique ID of the seller
Seller_Name	-	Seller's business name
Email	-	Seller's email address
phone	-	Seller's contact number

2.3 Product

Attribute	Key	Description
Product_ID	Primary Key (PK)	Unique ID of the product
Product_Name	-	Name of the musical instrument
Brand	-	Brand of the product (Yamaha, Fender, Casio, etc.)
Category	-	Product category (Guitar, Piano, Keyboard, Drums, etc.)
Price	-	Selling price
Stock	-	Available stock quantity
Description	Foreign Key (FK)	Product details and specifications

2.4 Order

Attribute	Key	Description
Order_ID	Primary Key (PK)	Unique ID of the order
Order_Date	-	Date the order was placed
Order_Status	-	Current status of the order
Total_Amount	Foreign Key (FK)	Total amount of the order

2.5 Payment

Attribute	Key	Description
Payment_ID	Primary Key (PK)	Unique ID of the payment
Payment_Method	-	Payment method (UPI, Credit Card, Debit Card, COD)
Payment_Status	-	Payment status
Payment_Date	-	Date of payment
Amount	-	Amount paid
Order_ID	Foreign Key (FK)	ID of the related order

2.6 Inventory

Attribute	Key	Description
Inventory_ID	Primary Key (PK)	Unique inventory ID
Product_ID	Foreign Key (FK)	ID of the product
Quantity_Available	-	Available quantity in stock
Last_update	-	date of last stock update

2.7 Review

Attribute	Key	Description
Product_ID Foreign Key (FK)	Foreign Key (FK)	Unique review ID
Rating	-	Customer rating (1–5 stars)
Comment	-	Customer feedback about the product
Review_Date	-	Date the review was submitted
Customer_ID	Foreign Key (FK)	ID of the customer who wrote the review
Product_ID	Foreign Key (FK)	ID of the reviewed product

2.8 Shipment

Attribute	Key	Description
Shipment_ID	Primary Key (PK)	Unique shipment ID
Order_ID	Foreign Key (FK)	ID of the related order
Courier_Name ☐	-	Name of the delivery service
Tracking_Number	-	Shipment tracking number
Delivery_Status	-	Current delivery status
Delivery_Date	-	Expected or actual delivery date

2.9 Relationship

Entity	Relationship	Related Entity	Cardinality
Customer	Places	Order	One Customer can place Many Orders
Seller	Sells	Product	One Seller can sell Many Products
Product	Stored In	Inventory	One Product has One Inventory Record
Order	Has	Payment	One Order has One Payment
Order	Has	Shipment	One Order has One Shipment
Customer	Writes	Review	One Customer can write Many Reviews
Product	Receives	Review	One Product can receive Many Reviews
Order	Contains	Product	One Order can contain Many Products, and One Product can belong to Many Orders